

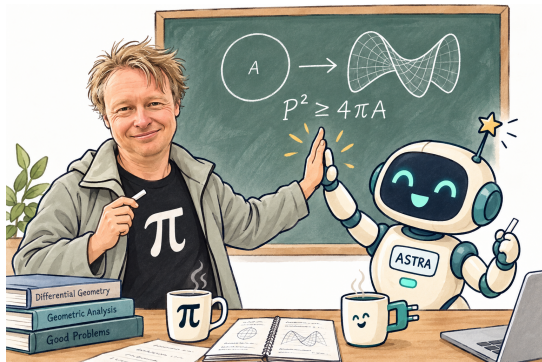
Isoperimetry and constant mean curvature

Glen Wheeler

University of Wollongong

AI and Tool Use Declaration.

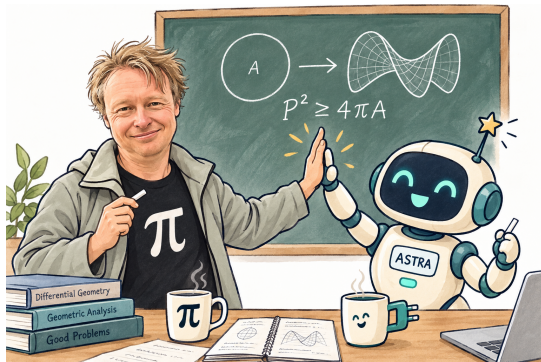
With ChatGPT 6.0 Astra



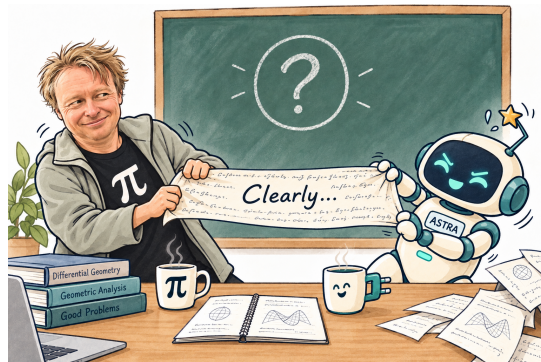
Making progress.

AI and Tool Use Declaration.

With ChatGPT 6.0 Astra



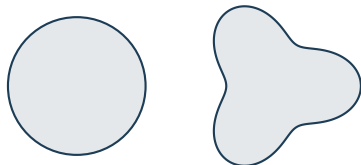
Making progress.



Not making progress.

How much boundary is needed?

The **isoperimetric problem**: enclose a prescribed volume using the least boundary area.



circle

another shape

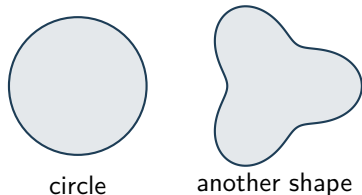
Same enclosed area.
Different boundary lengths.

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The **isoperimetric problem**: enclose a prescribed volume using the least boundary area.

In the plane, a circle is optimal:

$$P^2 \geq 4\pi V.$$



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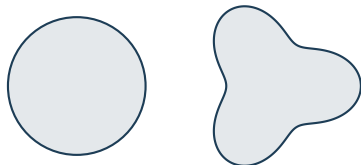
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In $\mathbb{R}^3$, a sphere is optimal:

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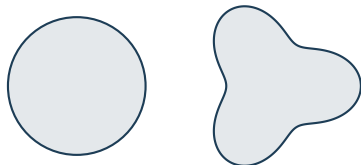
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Why should an optimal boundary have the same curvature everywhere?



circle

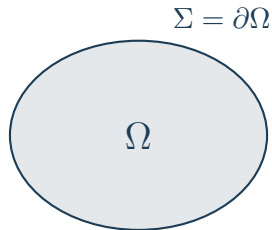
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Different boundary lengths.

A problem about shapes

Let $\Omega \subset \mathbb{R}^n$ be bounded, with smooth boundary
 $\Sigma = \partial\Omega$.

$$P = |\Sigma|, \quad V = |\Omega|.$$



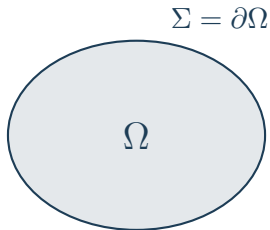
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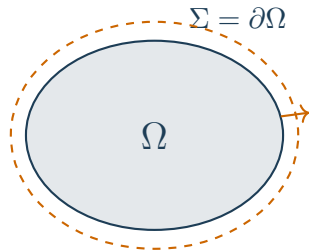
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Our variables are now shapes.

How do we take a derivative of a shape?



The calculus of variations



The calculus of variations

Leonhard Euler

1707–1783



The calculus of variations

Leonhard Euler

1707–1783

Euler developed systematic methods for finding extrema of quantities that depend on a whole curve.

His Methodus inveniendi appeared in 1744.

Instead of changing a number, we change a function.

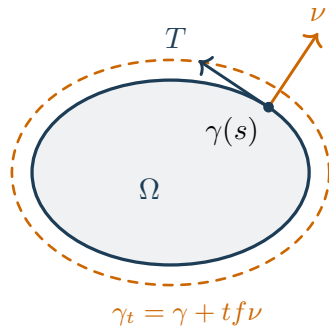


A small deformation of a curve

Parametrise a smooth closed plane curve by arclength:

$$\gamma = \gamma(s), \quad T = \gamma'(s), \quad |T| = 1.$$

Let ν be its **outward unit normal**.



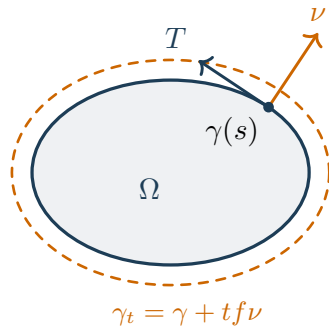
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$$\gamma_t(s) = \gamma(s) + tf(s)\nu(s).$$



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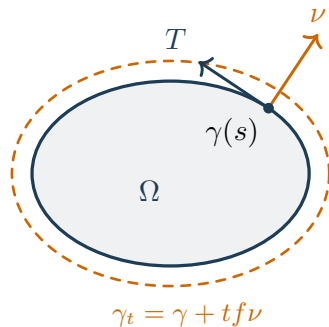
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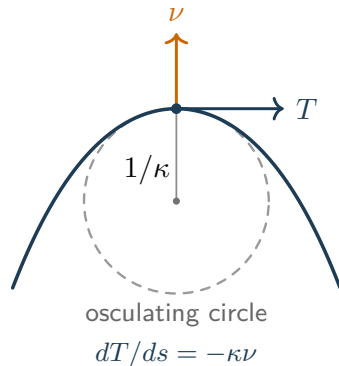
$f > 0$ moves outward; $f < 0$ moves inward.
A tangential velocity just changes the parametrisation to first order.



Curvature measures the turning of the tangent

With our choice of normal, define κ by

$$\frac{dT}{ds} = -\kappa\nu.$$



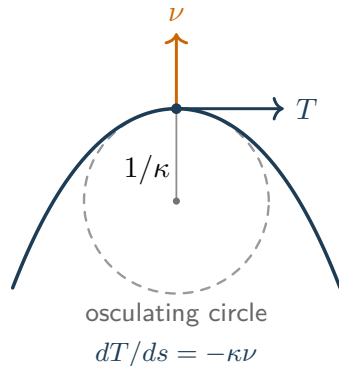
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Differentiating $\langle T, \nu \rangle = 0$ gives

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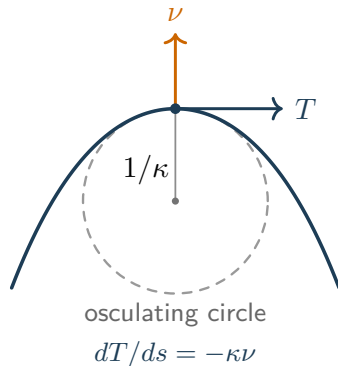
Differentiating $\langle T, \nu \rangle = 0$ gives

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For a circle of radius r ,

$$\kappa = \frac{1}{r} > 0.$$

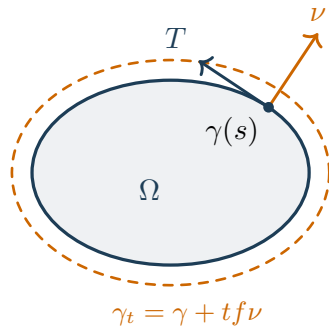
Tighter bending means larger curvature.



The first variation of length

Differentiate $\gamma_t = \gamma + tf\nu$ in s :

$$\partial_s \gamma_t = (1 + tf\kappa)T + tf'\nu.$$



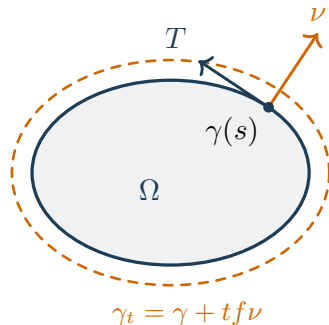
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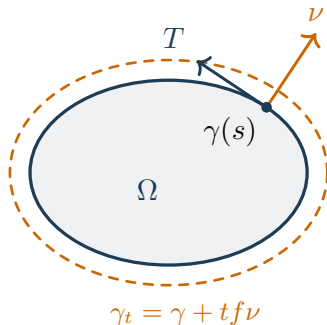
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Thus

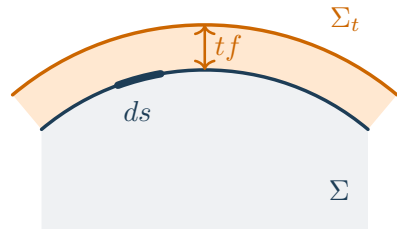
$$\left. \frac{d}{dt} \right|_0 |\partial_s \gamma_t| = \kappa f,$$

$$P'(0) = \int_{\Sigma} \kappa f \, ds.$$



The first variation of enclosed area

A short boundary segment of length ds sweeps out a thin strip.

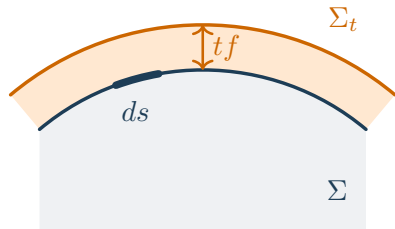


swept volume $\simeq tf ds$

The first variation of enclosed area

A short boundary segment of length ds sweeps out a thin strip.

$$\text{signed strip area} = tf \, ds + O(t^2) \, ds.$$



$$\text{swept volume} \simeq tf \, ds$$

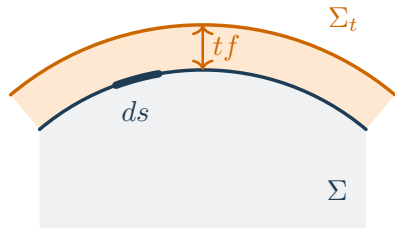
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Adding the strips gives

$$V'(0) = \int_{\Sigma} f \, ds.$$



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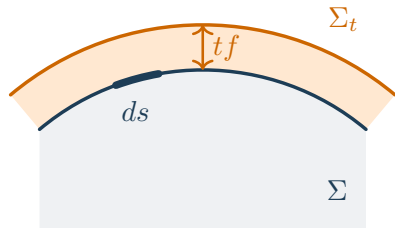
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Together:

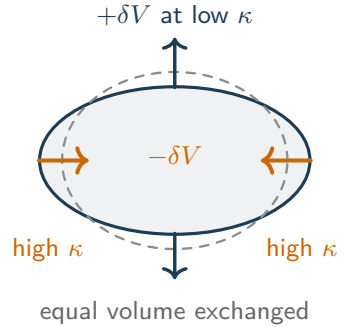
$$P'(0) = \int_{\Sigma} \kappa f \, ds, \quad V'(0) = \int_{\Sigma} f \, ds.$$



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Moving area between two boundary patches

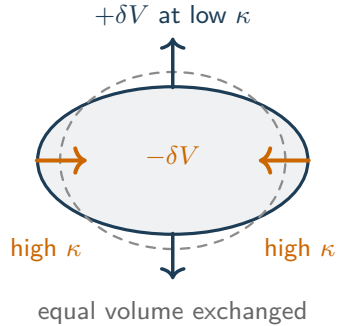
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Remove a little area at the more curved patch. Add the same amount at the flatter patch.

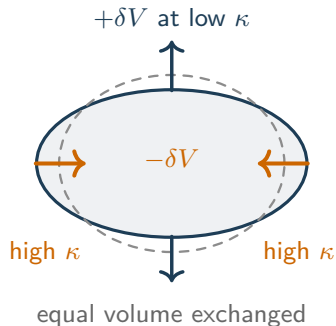


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$$\text{change in length} \approx (\kappa_{\text{low}} - \kappa_{\text{high}}) \delta V < 0.$$



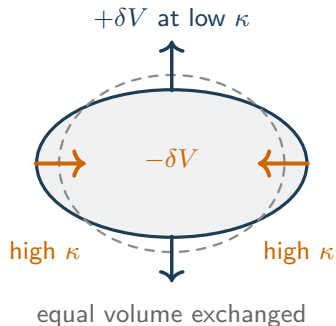
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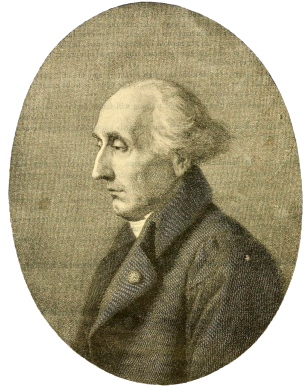
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At a minimiser, every patch must have the same marginal cost.



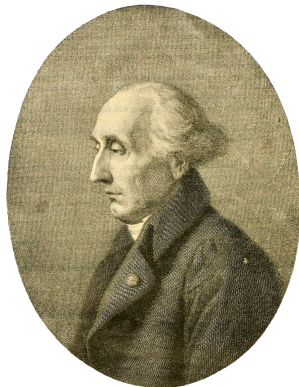
A constraint on the variation



A constraint on the variation

Joseph-Louis Lagrange

1736–1813



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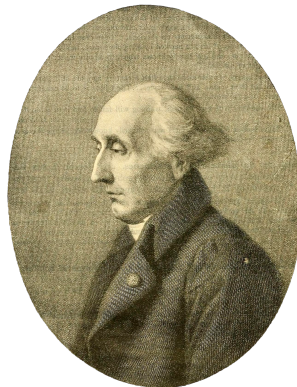
1736–1813

Lagrange developed the variational method and the use of multipliers for constraints.

For a fixed enclosed area, consider

$$P - \lambda V.$$

The multiplier will turn out to be curvature.

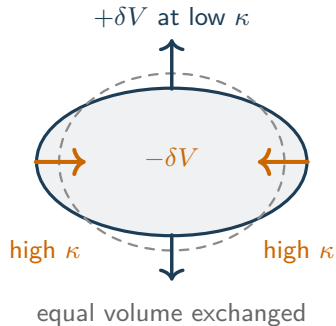


The Euler–Lagrange equation

At a constrained stationary curve, there is a constant λ such that

$$(P - \lambda V)'(0) = 0$$

for every smooth normal speed f .



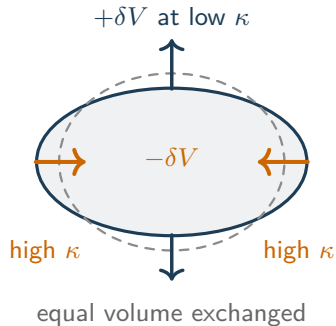
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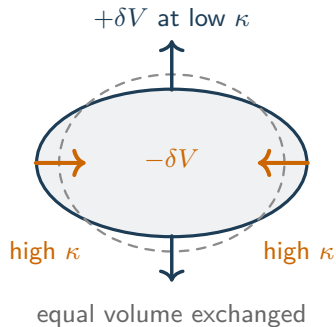
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Choose $f = \kappa - \lambda$:

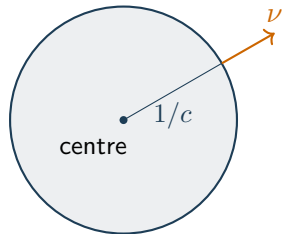
$$\int_{\Sigma} (\kappa - \lambda)^2 \, ds = 0.$$

Therefore $\kappa \equiv \lambda$.



Constant curvature gives a circle

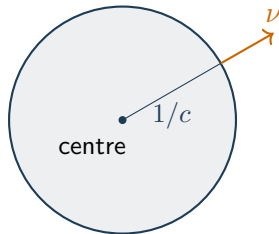
Suppose a connected closed plane curve has constant curvature $\kappa = c \neq 0$.



Constant curvature gives a circle

Suppose a connected closed plane curve has constant curvature $\kappa = c \neq 0$. Then

$$\frac{d}{ds} \left(\gamma - \frac{1}{c} \nu \right) = T - \frac{1}{c}(cT) = 0.$$



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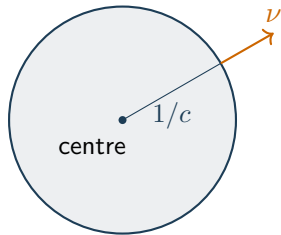
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So $\gamma - \nu/c$ is a fixed point, and

$$|\gamma - \text{centre}| = \frac{1}{|c|}.$$

The curve lies on a circle.



Constant curvature gives a circle

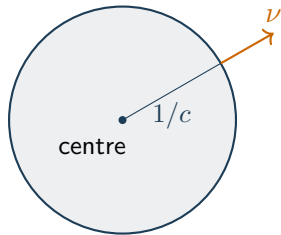
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The curve lies on a circle. This identifies a candidate. Global optimality still needs a proof.



Curvature of a surface



Curvature of a surface

Carl Friedrich Gauss

1777–1855



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Gauss developed the geometry of curved surfaces, relating their metric and curvature.

His Disquisitiones generales circa superficies curvas appeared in 1827.

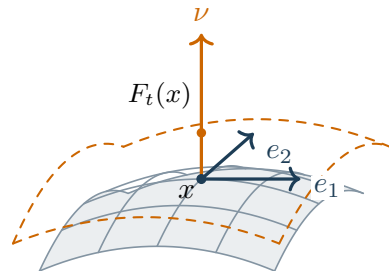
At a surface point, there is more than one direction in which to bend.



Principal curvatures

For a smooth hypersurface $\Sigma \subset \mathbb{R}^n$, the normal changes along tangent directions.

$$S(v) = D_v \nu.$$



two principal directions

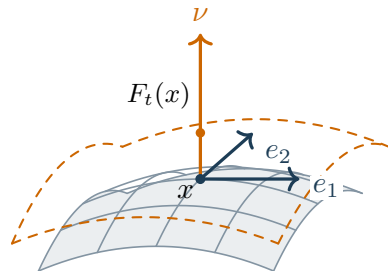
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The eigenvalues of S are the **principal curvatures**

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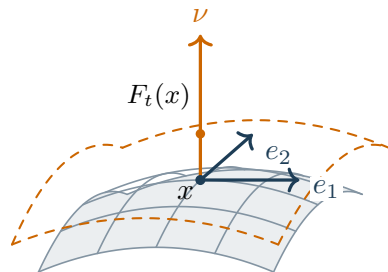
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We use **trace mean curvature**:

$$h = \operatorname{tr} S = \kappa_1 + \dots + \kappa_{n-1}.$$

Here “mean” means the sum, with no averaging factor.

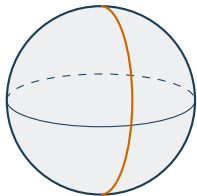


two principal directions

A sphere and a cylinder

For a radius- r sphere in $\mathbb{R}^3$,

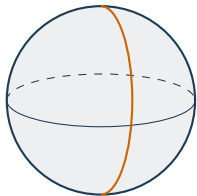
$$\kappa_1 = \kappa_2 = \frac{1}{r}, \quad h = \frac{2}{r}.$$



A sphere and a cylinder

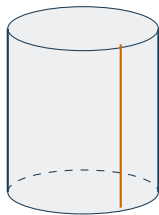
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For the side of a radius- r cylinder,

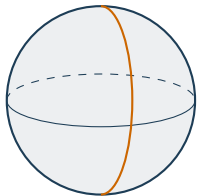
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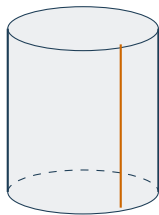
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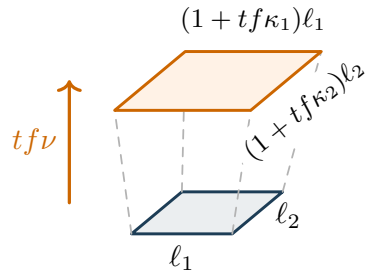
The Gaussian curvature is a different quantity: $K = \kappa_1 \kappa_2$.

How a surface patch changes area

Move Σ by $F_t(x) = x + tf(x)\nu(x)$.

In a unit principal direction e_i ,

$$dF_t(e_i) = (1 + tf\kappa_i)e_i + t(e_i f)\nu.$$



an infinitesimal area element

How a surface patch changes area

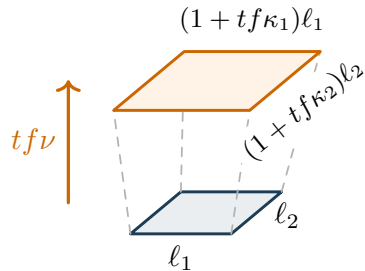
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$$\prod_{i=1}^{n-1} (1 + tf\kappa_i) = 1 + tfh + O(t^2).$$



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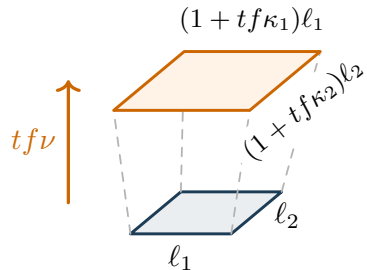
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Therefore

$$P'(0) = \int_{\Sigma} hf \, dA.$$



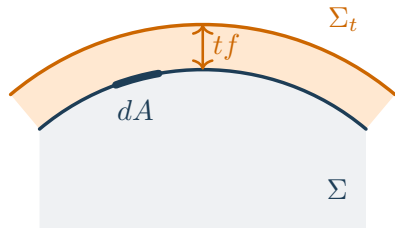
an infinitesimal area element

The two first variation formulas

For a closed smooth boundary in $\mathbb{R}^n$,

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$$V'(0) = \int_{\Sigma} f dA.$$



swept volume $\simeq tf dA$

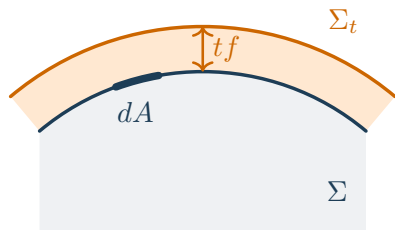
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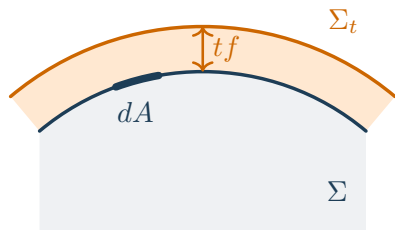
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The second formula again counts thin strips swept out by the boundary. **Mean curvature is the local rate of boundary area per unit added volume.**

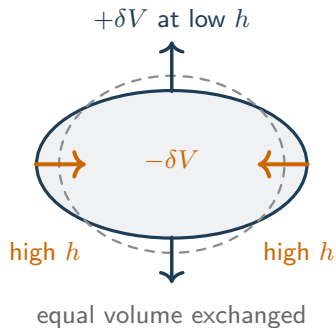


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A smooth minimiser has constant mean curvature

Let

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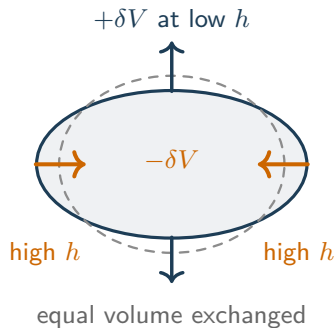
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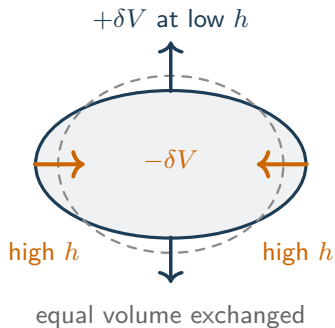
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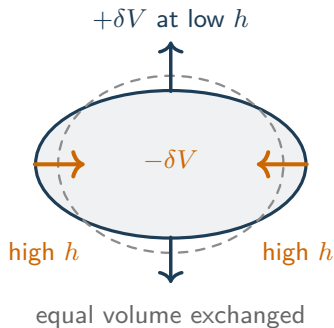
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Unless h is constant, perimeter decreases.

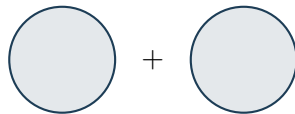


Stationary does not mean minimising

Take two disjoint balls of the same radius r in $\mathbb{R}^n$.

$$h \equiv \frac{n-1}{r}$$

on both boundary components.

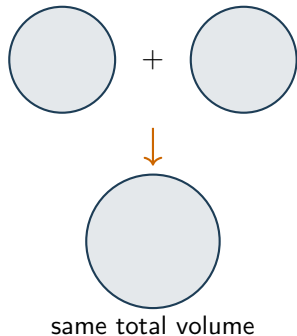


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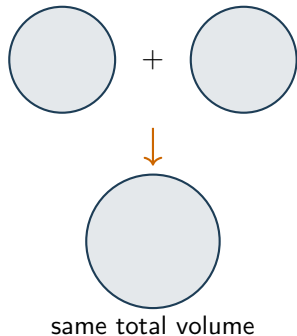
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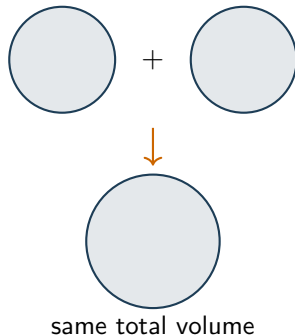
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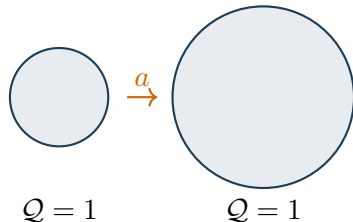
CMC is a **necessary condition** for a smooth minimiser.



A ratio that does not change under scaling

In $\mathbb{R}^n$, dilation by $a > 0$ gives

$$P \mapsto a^{n-1}P, \quad V \mapsto a^n V.$$



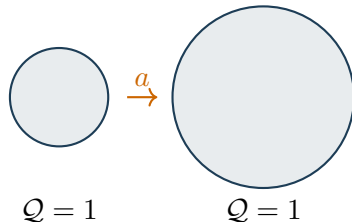
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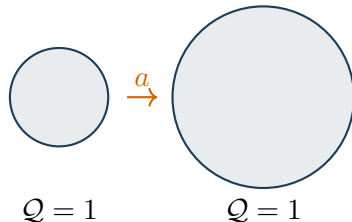
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Here ω_n is the volume of the unit ball.

Every Euclidean ball has $Q = 1$.

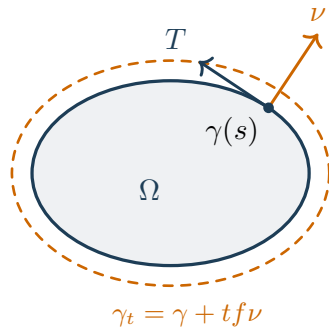
Fixed-volume minimisation and ratio minimisation are related by dilation.



Differentiating the isoperimetric ratio

Logarithmic differentiation gives

$$\frac{Q'(0)}{Q} = n \frac{P'(0)}{P} - (n-1) \frac{V'(0)}{V}.$$



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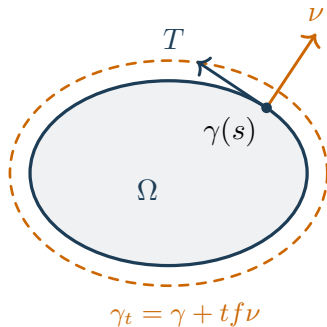
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Substitute the first variations:

$$\frac{Q'(0)}{Q} = \frac{n}{P} \int_{\Sigma} (h - H_0) f \, dA,$$

where

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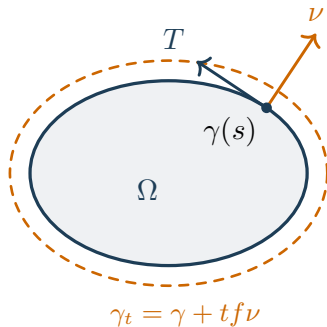
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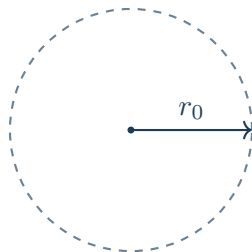
Stationarity gives $\boxed{h \equiv H_0}$.



The mean-curvature scale in the morning talk

Set

$$\theta = \mathcal{Q}(\Omega), \quad H_0 = \frac{(n-1)P}{nV}, \quad r_0 = \frac{n-1}{H_0}.$$



A comparison radius.
No spherical shape is assumed.

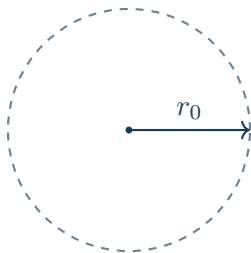
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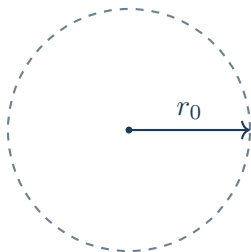
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Consequently

$$\begin{aligned} V &= \omega_n \theta r_0^n, \\ P &= n\omega_n \theta r_0^{n-1}. \end{aligned}$$



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Geometry determined by a metric



Geometry determined by a metric

Bernhard Riemann

1826–1866



Geometry determined by a metric

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Riemann's 1854 habilitation lecture introduced a framework for geometry in any dimension.

The metric tells us how to measure lengths and angles at each point.

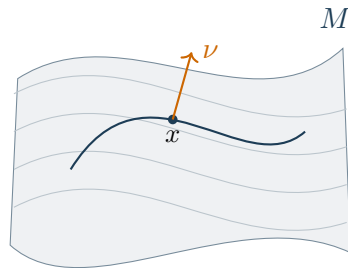
Can the same variation argument work in a curved ambient space?



Lengths, volumes and normals in a manifold

A Riemannian metric g_x is an inner product on each tangent space $T_x M$.

$$L(\gamma) = \int \sqrt{g_{\gamma(t)}(\dot{\gamma}, \dot{\gamma})} dt.$$

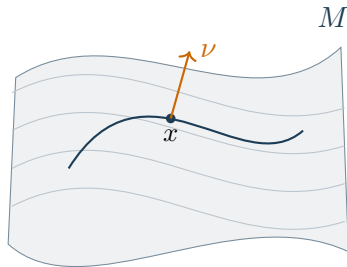


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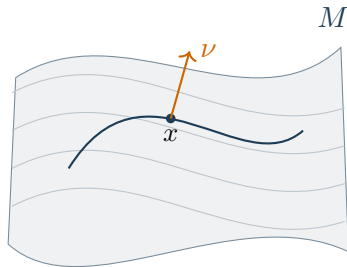
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It also gives volume dV_g , boundary area dA_g , and an outward unit normal ν . A geodesic is a locally straight path.

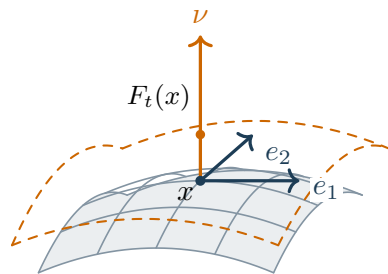
$\exp_x(v)$ follows the geodesic starting at x with initial velocity v for time one.



The same first variations in a curved space

Use geodesics for the normal deformation:

$$F_t(x) = \exp_x(tf(x)\nu(x)).$$



two principal directions

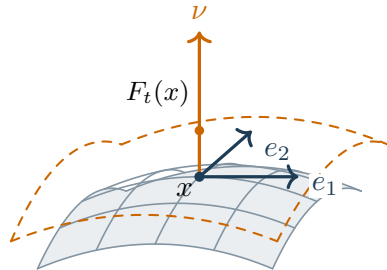
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Replace $D\nu$ by the covariant derivative:

$$h = \operatorname{div}_\Sigma \nu = \sum_{i=1}^{n-1} g(\nabla_{e_i} \nu, e_i).$$



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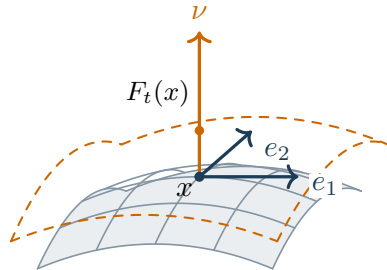
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The calculation still gives

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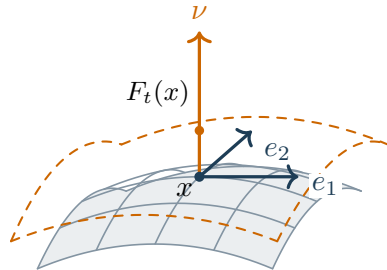
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So a smooth isoperimetric boundary still has **constant mean curvature**.



two principal directions

Geodesic circles in constant curvature

On a surface of constant curvature K ,

$$g = dr^2 + s_K(r)^2 d\varphi^2,$$

where

$$s_K(r) = \begin{cases} \sin r, & K = +1, \\ r, & K = 0, \\ \sinh r, & K = -1. \end{cases}$$

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$\sin r$

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r

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same initial angle and radial distance
transverse length = $s_K(r) d\theta$

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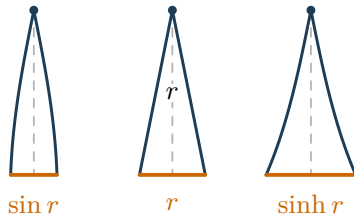
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$$P = 2\pi s_K(r), \quad V' = P.$$

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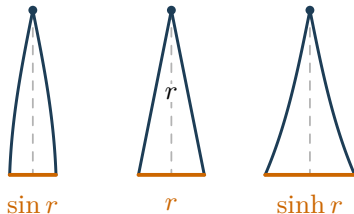
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$$\text{Thus } h = \frac{1}{P} \frac{dP}{dr} = \frac{s'_K(r)}{s_K(r)}.$$

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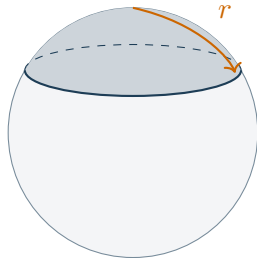


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A spherical cap separates the two minimisation problems

On the unit sphere, a cap of radius $0 < r < \pi$ has

$$V = 2\pi(1 - \cos r), \quad P = 2\pi \sin r.$$



The boundary moves
along the sphere.

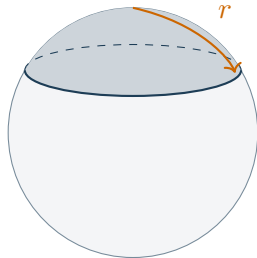
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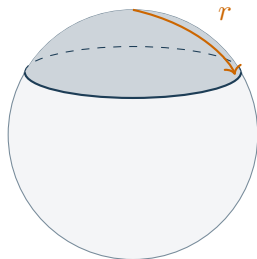
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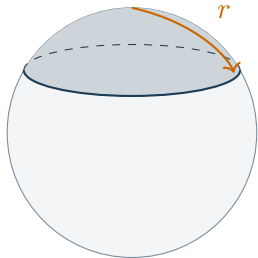
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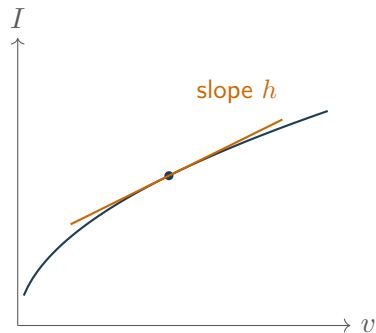


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Mean curvature is the slope of the isoperimetric profile

Let

$$I(v) = \inf\{P(\Omega) : V(\Omega) = v\}.$$



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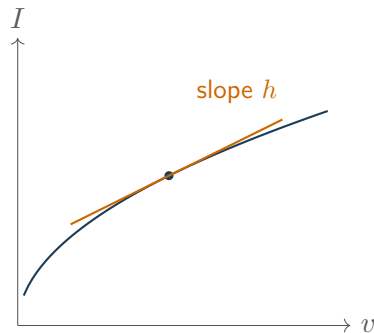
Let

$$I(v) = \inf\{P(\Omega) : V(\Omega) = v\}.$$

Suppose a smooth family of minimisers Ω_v exists near a volume v .

Differentiating $V(\Omega_v) = v$ gives

$$\int_{\partial\Omega_v} f dA_g = 1.$$



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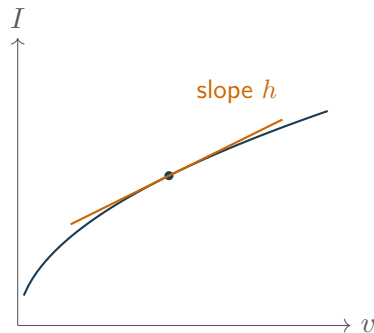
Suppose a smooth family of minimisers Ω_v exists near a volume v .

Differentiating $V(\Omega_v) = v$ gives

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Since h is constant on each boundary,

$$I'(v) = h.$$



Existence and smoothness are the natural next step

The calculation began with a smooth minimiser.



Bounded perimeter and volume
need not keep a region nearby.

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Does a minimiser exist?

On a closed manifold, compactness for sets of finite perimeter gives existence at each nontrivial volume.



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The regular part is smooth and CMC. Interior singularities can occur in ambient dimension $n \geq 8$.



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On a noncompact manifold, a minimising sequence may escape to infinity.

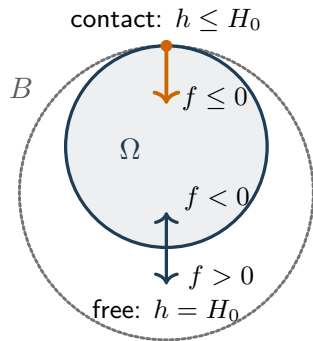


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The constrained minimiser in Proof part I

The morning talk minimises $\mathcal{Q}$ among regions confined to a ball B .

The perimeter counts the **whole boundary**, including contact with ∂B .



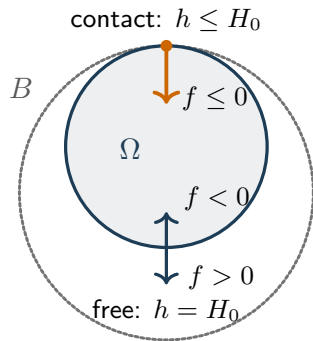
all variations stay inside B

The constrained minimiser in Proof part I

The morning talk minimises $\mathcal{Q}$ among regions confined to a ball B .

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The constrained minimiser in Proof part I

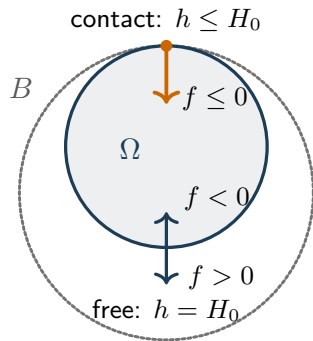
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At smooth contact, only inward variations are available:

$$f \leq 0, \quad \mathcal{Q}'(0) \geq 0 \quad \implies \quad h \leq H_0.$$



all variations stay inside B

The constrained minimiser in Proof part I

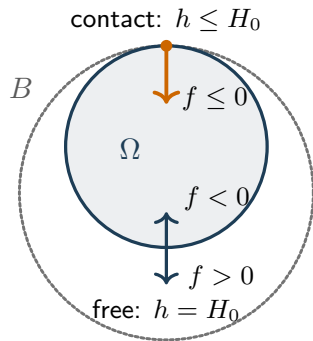
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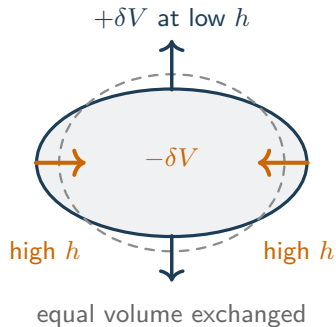
all variations stay inside B

Additional regularity and convex-obstacle arguments give the weak bound $|\mathbf{H}| \leq H_0$.

Isoperimetry and constant mean curvature: Recap!

The two derivatives are

$$P' = \int_{\Sigma} h f dA_g, \quad V' = \int_{\Sigma} f dA_g.$$



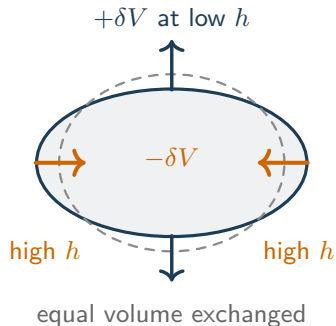
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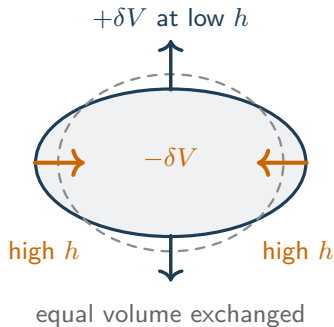
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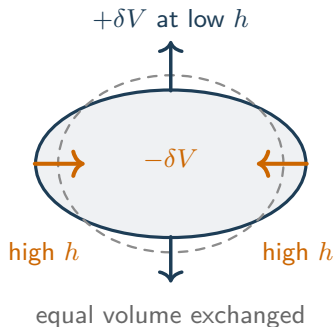
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A global optimisation problem imposes a local geometric condition because we can vary each small boundary patch.



Thank you for your attention. Any comments?

Further detail: what changes at second order?

At a CMC boundary $h = H_0$, a volume-preserving speed satisfies

$$\int_{\Sigma} f \, dA_g = 0.$$

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$$\int_{\Sigma} f \, dA_g = 0.$$

The second variation is

$$(P - H_0 V)''(0) = \int_{\Sigma} \left(|\nabla_{\Sigma} f|^2 - (|S|^2 + \text{Ric}_g(\nu, \nu)) f^2 \right) dA_g.$$

Further detail: what changes at second order?

At a CMC boundary $h = H_0$, a volume-preserving speed satisfies

$$\int_{\Sigma} f \, dA_g = 0.$$

The second variation is

$$(P - H_0 V)''(0) = \int_{\Sigma} \left(|\nabla_{\Sigma} f|^2 - (|S|^2 + \text{Ric}_g(\nu, \nu)) f^2 \right) dA_g.$$

A minimiser requires this to be nonnegative for every such f .

Ambient curvature now appears explicitly.

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